PROJECT REQUIREMENTS — DRIVEEASE

Project Name:
DriveEase — Online Car Rental Management System

Project Description:
DriveEase is a web-based car rental application that allows users to browse available cars, log in, and use booking features. Administrators can manage car listings and bookings.

TECHNOLOGIES USED:
- Frontend: React.js, Vite, JavaScript
- Backend: Node.js, Express.js
- Database: MongoDB Atlas
- Authentication: JSON Web Token (JWT)
- Frontend Hosting: Vercel
- Backend Hosting: Render
- Version Control: Git and GitHub

FUNCTIONAL REQUIREMENTS:
1. Users can register and log in.
2. Users can view available cars and their details.
3. The system displays car name, brand, category, year, seating capacity, transmission, fuel type, daily rental price, image, description, and availability.
4. Users can access the car booking features.
5. Administrators can add new car listings.
6. Administrators can update existing car details.
7. Administrators can delete car listings.
8. Administrators can view bookings and update booking status.
9. The frontend communicates with the backend through REST APIs.
10. The system stores application data in MongoDB Atlas.

SYSTEM REQUIREMENTS:
- A computer or mobile device
- A modern web browser
- Internet connection
- Node.js and npm for local development
- MongoDB Atlas account
- Git and GitHub account

DEPLOYMENT DETAILS:
Frontend URL:
https://drive-ease-weld.vercel.app/

Backend URL:
https://driveease-api-d1q6.onrender.com/

API Base URL:
https://driveease-api-d1q6.onrender.com/api

ENVIRONMENT VARIABLES:
Frontend:
VITE_API_URL=https://driveease-api-d1q6.onrender.com/api

Backend:
CLIENT_URL=https://drive-ease-weld.vercel.app

SECURITY REQUIREMENTS:
- Authentication must be handled by the backend.
- Administrative operations should be restricted to authorized users.
- MongoDB credentials and JWT secrets must remain private.
- Secret environment variables must not be exposed in frontend code or committed to GitHub.